APPLIED AI ENGINEER ASSESSMENT · CREDITFIX GROUP

AI-Enabled District Administration

A practical plan for clear, accountable citizen services

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Hypothetical scenario for a district administration

We recommend a phased change, with people in control

CORE STRATEGY

RECOMMENDATION

- Create one clear, traceable way to handle citizen requests.
- Automate simple, routine office handoffs and reminders.
- Use AI only for limited tasks: sorting requests, reading documents and finding information.
- Officers keep all official decisions.
- Test the value in a small pilot before scaling up.

NEXT STEPS

DECISION SOUGHT

Approve a discovery study of the real process and a small pilot, subject to administrative, security and procurement approvals.

Hypothetical district scenario. Study the real process before choosing any technology.

CURRENT ENVIRONMENT

In this hypothetical district, many linked services reach citizens

THE OPERATING CONTEXT

- Citizen complaints and service requests.
- Certificates and licences.
- Files moving between departments.
- Field inspections and site reports.
- Meeting actions and officer follow-ups.

Sample scenario (illustrative only, not a real district record): a resident reports a blocked drain near a school. The request may pass through several desks before a team visits the site.

CHANNELS USED IN THIS SCENARIO

Paper files | Phone calls | WhatsApp groups | Spreadsheets | Manual follow-ups



OPERATIONAL CHALLENGES

Fragmented processes may cause delay, rework and weak visibility

POTENTIAL BOTTLENECKS TO CHECK

THE CHALLENGE

01

Many ways to report

Requests may arrive by phone, email, web form or in person.

02

Unclear ownership

It may be unclear who owns a case during hand-offs.

03

Repeated data entry

Staff may type the same details into more than one record.

04

Repeated follow-ups

Citizens may need to call or visit again to ask for an update.

05

Limited visibility

Senior officers may lack one clear view from start to finish.

Sample scenario (illustrative only): A resident reports uncollected waste by phone. The report is typed again into a second system. The resident calls back for an update, and the officer cannot see its status.

Important: These are hypothetical issues to check during discovery, not proven findings. The sample is not an actual district record or measured case.

EXPECTED OUTCOMES

Measure success by outcomes for citizens and officers

TARGET STATE

GOALS TO CHECK IN DISCOVERY

1

Each request gets its own record that can be tracked.

2

Each open task has a named owner and a due date.

3

Staff can see each case's history and the next step.

4

Citizens are told their request was received and get updates.

5

Senior officers can spot old cases and blockages.

FRAMEWORK

GUIDING PRINCIPLES

CITIZEN-FIRST | OFFICERS STAY ACCOUNTABLE | SECURE BY DESIGN | SYSTEMS SHARE DATA | DELIVERED IN PHASES

Hypothetical district scenario. These goals are to be checked during discovery, not promised results.

DISCOVERY APPROACH

Study the real process before choosing technology

DISCOVERY STEPS (HYPOTHETICAL DISTRICT)

1. **Stakeholder interviews**
   Talk to officers, clerks, field staff, citizens, and IT and security teams.
2. **Service observation**
   Watch selected services from start to finish.
3. **Case tracing**
   Follow sample cases across departments and handovers.
4. **Artefact review**
   Review forms, logbooks, spreadsheets and current software.
5. **Constraint mapping**
   Map exceptions, approvals, service rules and data limits.

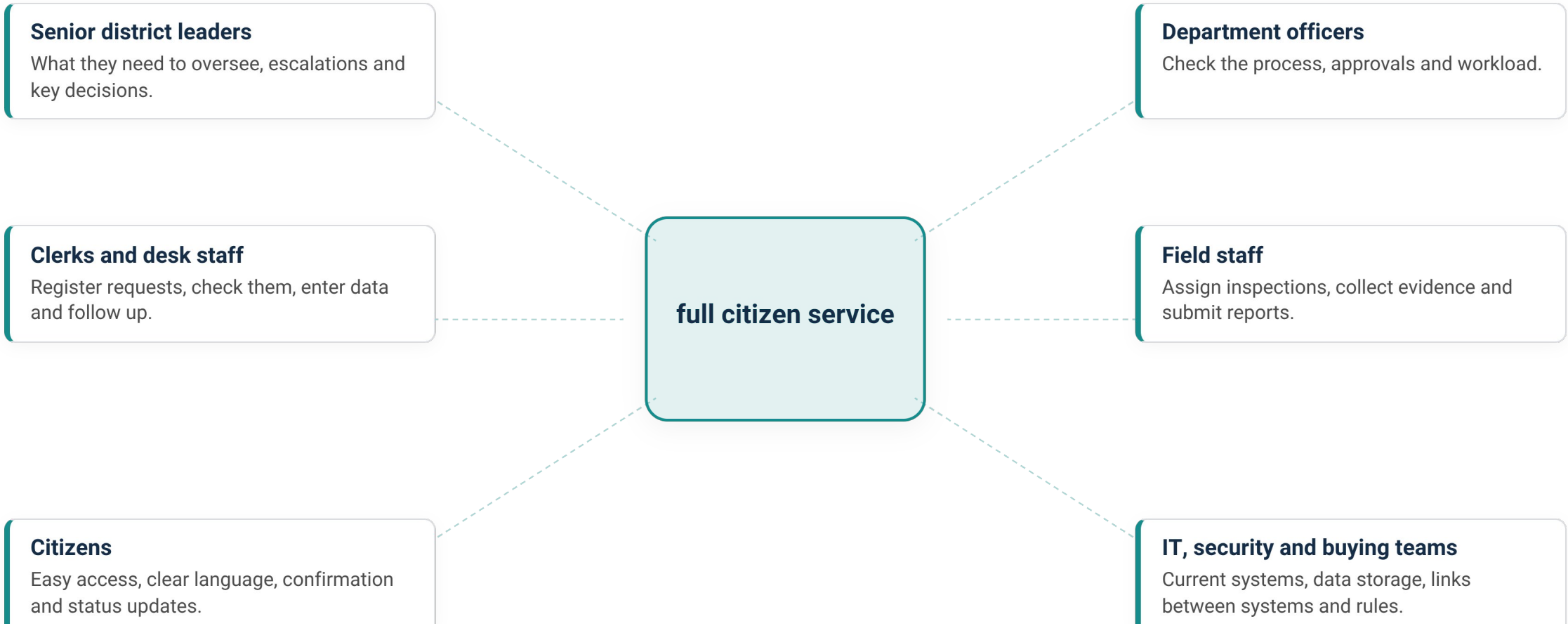
DISCOVERY OUTPUTS

A checked map of today's process | Main pain points, ranked | Starting measures | Clear requirements | A pilot recommendation

STAKEHOLDERS

Each stakeholder sees a different part of the service

HYPOTHETICAL DISTRICT: WHO WE NEED TO HEAR FROM



DISCOVERY QUESTIONS

Interviews show needs, limits and workarounds

HYPOTHETICAL DISTRICT · INTERVIEWS

SAMPLE INTERVIEW QUESTIONS

LEADERSHIP

SENIOR OFFICERS

- How do you find overdue cases today?
- Which reports are hard to prepare?

OPERATIONS

DEPARTMENT STAFF

- How are requests received, checked and assigned?
- Where do files wait or get sent back?

FIELD CHECKS

FIELD STAFF

- How are visits, proof and reports recorded?

PUBLIC

CITIZENS

- How do you send a request and check its status?

RULES & SYSTEMS

IT / SECURITY

- Which systems and approved ways to connect already exist?
- What rules apply to data storage and access?

CURRENT WORKFLOW

Current process: seven steps, with waits and hand-offs

Illustrative current-state process, to validate during discovery.

01	02	03	04	05	06	07
Submit	**Receive**	**Register**	**Route**	**Act**	**Respond**	**Close**
Citizen submits a request	Office receives the request	Clerk registers and checks documents	Request is sent to a department	Officer reviews and takes action	Decision or response is sent to the citizen	Case is closed, or returned for more information

Waiting point

Time from submission to first action

Hand-off 1

Citizen to office

Waiting point

Documents wait for a clerk check

Hand-off 2

Office to department

Waiting point

Officer queue before action

Hand-off 3

Decision back to citizen

Waiting point

Requests wait for more information

EXAMPLE (ILLUSTRATIVE)

“The drain outside our street is overflowing.” It is registered, routed to drainage, and may wait in an officer queue.

VALIDATION NOTE

Confirm these steps, hand-offs and waits with the office during discovery.

BASELINE ANALYSIS

Measure the current situation before building anything

HOW TO INVESTIGATE

- Sample cases across their full lifecycle.
- Time each case spends at every stage.
- Count hand-offs between people and departments.
- Track repeat visits and enquiries about the same issue.
- Record rework, returns and missing documents.
- Observe manual registers, calls and follow-up routines.

BASELINE MEASURES

Measures to collect:

Time at each stage	**Hand-offs per case**
Repeat visits and enquiries	**Rework and returns**
Backlog and ageing	**Staff effort**

No baseline figures have been collected yet. None are assumed. Collect and agree them with the office during discovery.

USE CASE PRIORITIZATION

Use cases can be compared by public value, ease and risk

Initial hypothesis to validate, not a measured ranking. Horizontal axis: ease of implementation. Vertical axis: expected public value.

HARDER TO BUILD · HIGHER VALUE

- AI assistant for officers
- Multilingual citizen assistant

EASIER TO BUILD · HIGHER VALUE

- Unified citizen intake
- AI grievance classification
- Rules-based department routing
- Digital case tracking

HARDER TO BUILD · LOWER VALUE

- Document intelligence

EASIER TO BUILD · LOWER VALUE

- Field report and meeting actions
- Officer dashboard

Risk check: security, legal and AI risk is assessed for each use case before any pilot. Proposed pilot to validate: intake, classification assistance, rules-based routing and case tracking.

PROPOSED USE CASES

Nine use cases: only some of them use AI

AI role states whether AI is used. “No AI” means standard software or approved rules. Examples are illustrative, not district records.

1. Unified citizen intake

AI role: optional speech-to-text or language support. Case ID and storage are standard software.

Example: a spoken drain complaint is turned into text for staff to review.

2. Grievance classification

AI role: suggests category, summary and possible priority for staff review.

Example: “Waste has not been collected for three days” is suggested as waste management.

3. Department routing

No AI: approved rules use the confirmed category, ward and department.

Example: a verified drainage complaint in Ward 12 goes to Ward 12’s drainage department.

4. Case tracking

No AI: workflow software stores owner, status, due date, history and reminders.

Example: a citizen sees a streetlight complaint is awaiting inspection.

5. Field reports and meeting actions

AI role: optional. Summarises notes and suggests actions; an officer approves them.

Example: a blocked-drain inspection note becomes a draft follow-up task.

6. Officer dashboard

No AI: standard database queries and charts show open, ageing and overdue cases.

Example: an officer sees which department has the most overdue complaints.

7. Document intelligence

AI role: reads scanned documents, extracts fields and drafts a summary. Staff verify.

Example: applicant name and application number are extracted from a scanned form.

8. Officer assistant

AI role: searches approved circulars, policies and SOPs, and answers with a source reference.

Example: an officer asks which documents a certificate process needs.

9. Citizen assistant

AI role: answers common questions from approved public information, in validated languages.

Example: a citizen asks how to report a water-supply problem and gets the official steps.

• UNIFIED CITIZEN INTAKE

Unified intake creates one traceable front door

Citizens can submit online, and staff can register requests for people who visit or call.

HOW IT WORKS

- Online submission and staff-assisted registration use the same intake.
- Each request gets one case ID.
- The request and its attachments are stored in one place.

Example: “The streetlight near the school is not working.” (illustrative)

STANDARD SOFTWARE OR AI?

Task	Type
Case registration	Standard software
Case ID generation	Standard software
Storing the request	Standard software
Voice message to text	Optional AI, staff check the text

AI is optional here. It only helps transcribe or understand a voice message. Staff review the result before it is used.

• AI GRIEVANCE CLASSIFICATION

AI suggests a category; a person confirms it

COMPLAINT TEXT

“The drain outside our street is overflowing.”

Illustrative example, not a district record.

AI SUGGESTS

- Category: Drainage
- Summary: Overflowing drain on a residential street
- Possible priority: High, for staff review
- Reason cues: the words “drain” and “overflowing”

OFFICER ACTIONS

- Confirm the suggested category
- Correct it if it is wrong
- Check and set the priority
- Record the final decision

AI suggests; a person confirms. The reason cues show the words behind each suggestion. Another example: “Waste has not been collected for three days” could be suggested as waste management (illustrative).

• AUTOMATED DEPARTMENT ROUTING

Approved rules route each case, not AI

Routing uses the confirmed category, the ward and an approved department map. AI does not choose the department.

Confirmed category

Set by a person

Ward or jurisdiction

Checked against the address

Approved department map

Category and ward mapped to a department

Department queue

The case enters the queue of the responsible department

EXAMPLE

A verified drainage complaint for Ward 12 goes to the department responsible for drainage in Ward 12.

EXCEPTION PATH

If the category or ward is unclear, the case goes to a manual check queue. A staff member decides the routing and records the reason.

The rules are approved in advance, and officers can see which rule sent each case.

• DIGITAL APPLICATION TRACKING

A shared case record gives every team the same status

This is workflow software, not an AI function.

Case ID

Unique number for each request

Owner

Person responsible now

Status

For example: assigned, awaiting inspection

Due date

When action is expected

History

Each step, with date and person

Reminders

Sent to the owner before a due date

Citizen updates

Status the citizen can see

EXAMPLE

A streetlight complaint is assigned to the electrical team with the status “Awaiting inspection”. The citizen sees the same update.

• FIELD REPORTS AND MEETING ACTIONS

Turn field reports and meeting notes into clear tasks

How AI helps

- AI reads meeting notes or an inspection report.
- It writes a short summary.
- It suggests action items, owners and due dates.

→

Human approval

- An authorised officer checks and approves each task.
- AI cannot send field staff or make official decisions.
- Approved tasks go into one task register.

Example: a roadside drain is reported blocked. AI drafts a summary and a follow-up task. The officer confirms the task before anyone is sent.

• OFFICER DASHBOARD

A dashboard shows case numbers so managers can act early

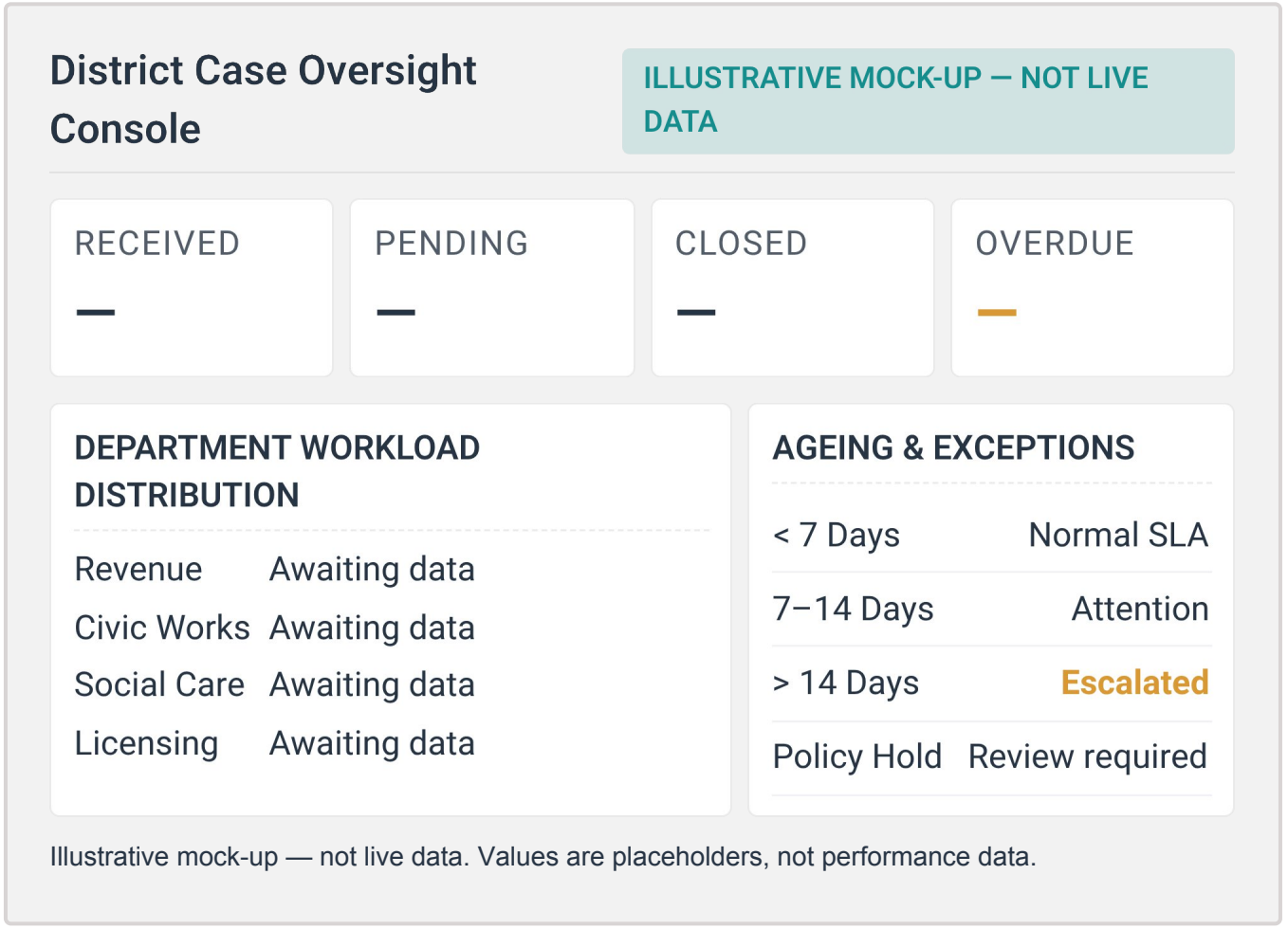
CORE VIEWS

- Requests received, pending and closed.
- Department-wise workload.
- How long each case has been open, and which are overdue.
- Cases escalated or needing attention.
- How workload changes over time.
- Open case details, for authorised staff only.

REPORTING

Standard software and database calculations build these numbers from checked case data. No AI is used.

Example: a senior officer sees several open water-supply complaints in one area and can review them together.



• DOCUMENT INTELLIGENCE

Document AI reads files faster, and staff check every result

Read the file

OCR (optical character recognition) turns a scan or upload into text.

Identify the type

The tool recognises the document, such as an application form.

Extract and summarise

It pulls out key details and writes a short summary.

Staff verify

Staff check each detail against the original file.

Example: from a scanned form, the tool proposes the applicant name, application number and date. Staff compare each one with the scan.

CONTROL

Poor scans and unclear handwriting must be checked by a person. Never treat extracted content as verified until staff have checked it.

• AI ASSISTANTS

Two assistants find approved information. Neither makes decisions.

Languages: Hindi and English at launch. More district languages will be confirmed during discovery.

A · INTERNAL

Officer assistant

- Searches only approved circulars, policies, service rules and procedures.
- Gives a short answer with links to its source.
- Example: a certificate-process checklist lists each document needed, with the rule it comes from.

B · PUBLIC-FACING

Citizen assistant

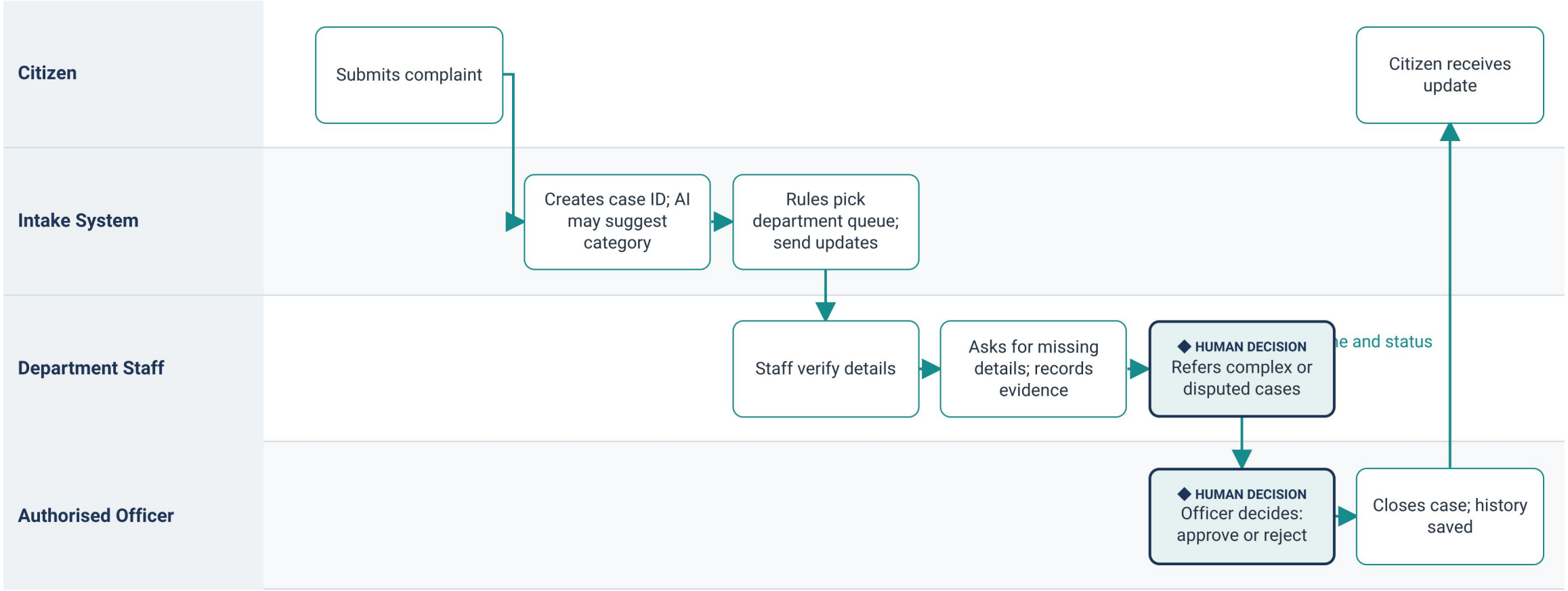
- Answers common questions from approved public information.
- Explains steps and where to submit. Example: a broken streetlight gets the official steps and the complaint route.
- Shows its sources. Checks case status only after secure verification.

⚠ GUARDRAILS

Assistants retrieve and explain information only. They do not approve applications, decide eligibility or replace officers. They respect user permissions and pass uncertain or case-specific questions to staff.

FUTURE WORKFLOW

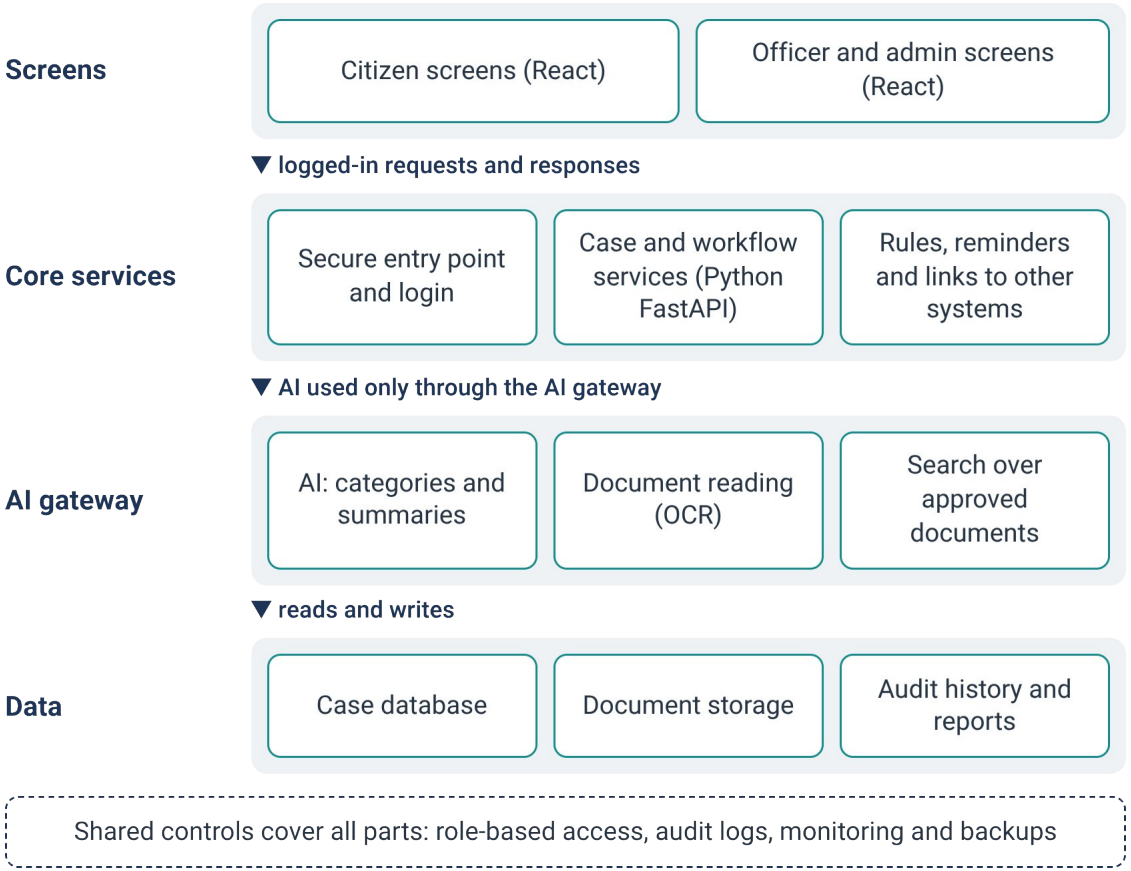
The future workflow makes each hand-off visible



Proposed, not live. ◆ marks a human decision. AI only suggests, for example a category or summary. Software rules route the case; people decide.

SYSTEM ARCHITECTURE

The system uses separate parts for screens, workflow, data and AI



Indicative technology mapping

Layer	Indicative technology
Web portal and dashboard	React
Backend and workflow APIs	Python FastAPI
Case and workflow database	PostgreSQL
Document storage	Object storage
Background jobs	Background job queue for OCR and reminders
AI services	Approved LLM and OCR services behind an AI gateway
Document search (only if needed)	pgvector or equivalent
Identity and access	Approved login provider with role-based access
Monitoring and audit	Monitoring and audit logging

Proposed, not live. Final choices depend on discovery, current government systems, hosting and security requirements. Messages go through approved email, SMS and WhatsApp providers.

END-TO-END DATA FLOW

Every AI-assisted step can be traced from start to finish

1 · INPUT

Citizen submits a request or document, for example an overflowing drain.

2 · RULE

Software checks required fields and creates a case ID.

3 · SYSTEM OF RECORD

Case and documents are stored securely.

4 · RULE

Approved rules route the case to a department and set due dates.

5 · AI SUGGESTION

AI helps only with language or documents, for example a summary. Its output is a suggestion.

6 · RULE

Software checks the AI output before staff see it.

7 · HUMAN REVIEW

Staff confirm or correct the suggestion.

8 · SYSTEM OF RECORD

The case is updated only after an authorised officer decides.

9 · AUDIT EVENT

Audit history records the input, AI suggestion, rule result, staff correction and final decision.

10 · RULE

Citizen gets an update; staff get reminders.

Legend: Rule · AI suggestion · Human review · System of record · Audit event

AI output is never a final action: only an authorised human decision updates the record.

AI VS. RULES VS. HUMAN DECISIONS

AI reads and drafts, rules apply fixed steps, and people decide

AI	RULES AND SOFTWARE	HUMAN OFFICERS
• Understands a free-text complaint, for example a report of an overflowing drain. • Summarises an inspection note. • Reads a scanned form. • Searches approved documents for a plain-language question.	• Creates a case ID and checks required fields. • Maps a confirmed category and ward to a department. • Calculates due dates and sends reminders. • Updates status and calculates dashboard numbers.	• Confirms uncertain information. • Closes sensitive or disputed cases. • Resolves exceptions and decides enforcement, eligibility and approvals.

AI suggests. Rules apply fixed steps. People make official decisions.

PROTOTYPE DEMONSTRATION

Illustrative Prototype Scope

Test the riskiest step first: AI complaint classification and routing.

Citizen submits a complaint.

AI suggests a category and summary.

Approved rules find the department, using category and ward.

Officer approves or corrects the result.

System saves the case, status and audit history.

SAMPLE COMPLAINT (SYNTHETIC)

“The drain outside our street is overflowing.”

WHAT AI DOES

AI helps understand the complaint and drafts the summary. Case ID, department mapping and the final decision are not AI.

ACCEPTANCE CHECKS

Workflow runs end to end and can be tested. It states the limits of the AI suggestion.

SECURITY AND PRIVACY

Security and privacy designed in from day one

Who can access data

- Each user gets only the access their role needs.
- Strong login for all staff accounts.

Protecting data

- Data is encrypted in transit and at rest.
- We collect only the data we need.
- Clear rules for how long data is kept and when it is deleted.

Monitoring and hosting

- Audit logs record access and important actions.
- Backups are secure and tested.
- Hosting is chosen to meet the authority’s requirements.
- Vendors and integrations are reviewed before go-live.

Governance

The relevant authority must confirm applicable Indian data protection law, government security requirements, and departmental policies.

AI RISK MANAGEMENT

Human review and technical controls manage AI failure modes

AI risk	Technical control	Human fallback	Control owner
AI invents an answer	Use approved sources only. Require source references.	Officer checks the cited source before acting.	AI product owner
Wrong complaint category	Confidence thresholds, test sets and monitoring.	Staff correct the category. Corrections are logged.	Service owner
Routing rule is wrong	Fixed mapping rules and exception queues.	Duty officer clears the exception queue.	Department routing lead
Language is misunderstood	Test representative languages before pilot use.	Staff handle unclear or non-English cases.	QA lead
Malicious input tries to expose data	Treat user input as untrusted. Limit tools. Check access outside the model.	Security incident process with a named responder.	Security lead
Results are unfair	Test across categories and languages. Monitor outcomes.	Appeal and correction route for every citizen.	Service owner with equality lead

USER ADOPTION

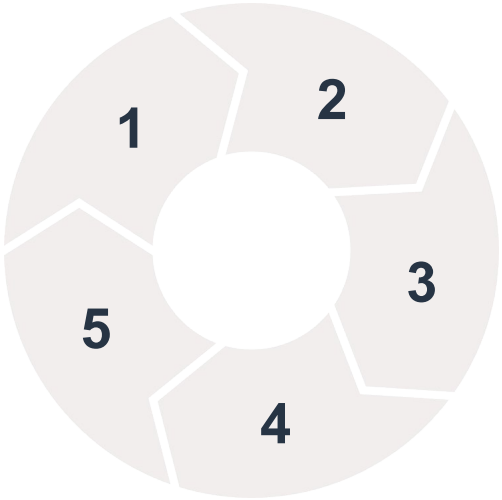
CHANGE AND ADOPTION

People must help shape and learn the new workflow

Adoption cycle

Co-design

Involve clerks and officers in designing the workflow.



Improvement

Act on feedback and refine the workflow.

Role-based training

Train each role on its own tasks, with quick guides.

Assisted support

Offer help desk and assisted routes for users.

Feedback

Collect user feedback and usability issues.

Adoption measures

Track active use, task completion, support requests and user feedback.

A manual or assisted route stays available whenever a user needs it.

PILOT SCOPE

A limited pilot should prove the workflow before expansion

In scope

- One high-volume grievance workflow.
- One or two departments, if agreed.
- Intake, case tracking and notifications.
- AI suggests complaint categories. Officers review them.
- Simple rules route cases. Officers use a dashboard.
- Audit trail for every case.

Out of scope

- District-wide rollout.
- Integrations not yet agreed.
- Autonomous decisions without officer review.
- Scale and ongoing support before a Go decision.

Go / No-Go criteria

- The workflow works from start to finish.
- Staff can run it without help.
- Security checks pass.
- Baseline and pilot measures reviewed.

Live data: can be used only after approval. Security and privacy controls must be in place first.

IMPLEMENTATION ROADMAP

Phased delivery plan from discovery to pilot evaluation

Planning estimates to validate after discovery. Not a quote or commitment.

Phase	Elapsed time	Indicative effort	Key deliverable	Roles
1. Discovery	1–2 weeks	10–15 person-days	Process maps, baseline, constraints and approvals plan	Project lead / business analyst; departmental champions
2. Design and build	4–6 weeks	45–65 person-days	Prototype, workflow configuration, security design and tested build	Full-stack engineers; AI engineer; UX; QA, DevOps and security
3. Controlled pilot	2–3 weeks	15–25 person-days	Deployment to agreed users, training and support log	Project lead; departmental champions; DevOps and support
4. Evaluation	1–2 weeks	5–10 person-days	KPI report, feedback review and Go / No-Go recommendation	Business analyst; QA; service owners
Total (phases 1–4)	9–14 weeks	75–115 person-days	—	—

A person-day is one person’s work for one day. Phases may overlap and depend on timely access and approvals. Excludes major legacy integrations, procurement lead time and district-wide rollout. Scale and ongoing support are outside this estimate and happen only after a Go decision.

SUCCESS METRICS

Success measures, baselined before the pilot

Measure	Definition	Baseline	Pilot	Target
Correct first-time routing	Share of complaints sent to the right department on the first try			
Time to acknowledge	Time from submission to acknowledgement			
Time to resolve	Time from submission to closure			
Open cases: number and age	How many cases are open, and how long they have been open			
AI suggestions accepted or corrected	Share of AI suggestions that officers accept or correct			
Staff use	Number of active users and completed workflow actions			
Citizen feedback	Score from feedback collected after each service			

Capture a pre-pilot baseline and set numerical targets only once the baseline is known.

FINAL RECOMMENDATION

Approve discovery first, then a limited pilot

Pilot-first commercial structure

1. Discovery and workflow assessment.
2. Fixed-scope pilot with agreed deliverables.
3. Evaluate against KPIs approved jointly.
4. Expand only if the pilot proves value.

Implementation, integrations, hosting and support are priced separately, after requirements are validated.

Objections answered

“We already have a system.” Integrate with and improve existing systems where practical.

“It is not secure.” Agree security, hosting and access controls before any live data is used.

“Our staff will not use it.” Co-design with staff, train by role and keep assisted routes.

DECISION REQUESTED

Approve discovery now. Approve a limited pilot only if discovery, security, administrative and procurement checks support it.

Make every request traceable. Make every task accountable. Keep every decision human-led.